Iterated Symmetric Centipede: Learning and Evolutionary Games

I. Lazaridis and Athanasios Kehagias

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Abstract

We study the *Iterated Symmetric Centipede* (ISC) game. The ISC involves repeated plays of the *One-shot Symmetric Centipede* (OSC), which is the "classic" Centipede game with random selection of the first moving player. Since ISC involves repeated plays of OSC, we can use *history-dependent* strategies, similar to the All-C, All-D, Grim etc. used in the *Iterated Prisoner's Dilemma*. We are especially interested in the interaction between these strategies. We first study the Nash equilibria of the ISC using standard game theoretic methods. Then we consider an *evolutionary game* using the ISC as the "base game". We study the evolutionary game for (i) the case of large populations, using *replicator dynamics* and (ii) for the case of *small populations* using Markov chains (obtained from the use of *pairwise proportional imitation*). We see that the presence of cooperation-promoting strategies (such as Grim) favors cooperation. A *learning theory* interpretation of replicator dynamics and pairwise proprtional imitation is also presented.

1 Introduction

In this paper we study games derived from the *Centipede game*, an extensive form game of perfect information. The "classic" Centipede game was introduced in [15]; a recent overview of *experimental* approaches to the field appears in [10]. In the current paper we study normal form variants, especially the *iterated symmetric centipede* (ISC), which is obtained by the following procedure.

First, using a standard construction, we obtain a normal form game which is equivalent to the "classical" (tree) Centipede. To remove the asymmetry between first and second player of the original Centipede we modify the game by randomizing the selection of the player who has the first move; in this way we obtain the one-round *symmetric* centipede (OSC). This has been previously studied by several authors; e.g., see [3, p.238, pp.255-260] and [4, 5] for a general analysis, [17, 8, 14] for an evolutionary games approach and [11] for the use of genetic algorithms.

However, as mentioned, we are mainly interested in the iterated symmetric centipede (ISC) game, which consists of playing T rounds of the OSC. We assume that, in every round of ISC, both players recall the moves and outcome of the previous rounds. This introduces the possibility of using *history-dependent* strategies; our main interest in the current paper is the effect of using such strategies to promote cooperation and punish defection.

Hence we adopt an interpretation of repeated Centipede play at the *round-level*, where each round consists of a complete one-shot Centipede "tournament" and strategies update only *between rounds*. Under this interpretation it is natural to lift strategies from the *Iterated Prisoner's Dilemma* (IPD) context, such as Tit-for-Tat and Grim Trigger. To produce the current-round behavior, such strategies use the opponent's actions in previous rounds (rather than simply reacting to within-round decisions). In short, the subject of our paper can be understood as an *evolutionary tournament* of the type previously used to study the IPD (most famously by Axelrod in [1]).

As far as we can tell, this approach has not been previously reported in the literature. The closest we have found is several works which apply various learning and / or evolutionary frameworks to the Centipede [8, 12, 13, 17, 18, 19, 20]. For example, in [20] Centipede play is modeled as a stochastic learning process in which players update strategies between rounds based on opponents' prior behavior; these learning dynamics capture adaptive round-level responses consistent with Tit-for-Tat–type reciprocal strategies, even when players do not follow deterministic trigger rules. The other cited papers follow similar approaches, usually applied to the extensive form (and often *un*symmetric) Centipede.

The current paper is structured as follows. In Section 2 we briefly review the "classic" (tree) Centipede and the normal form one-shot symmetric Centipede. In Section 3 we present the iterated symmetric centipede game (ISC). In Section 4 we identify the Nash equilibria (NE) of ISC, in Section 5 we study the replicator dynamics of an evolutionary game with ISC as the base game and in Section 6 we study the evolutionary stable strategies. In Section 7 we present a Markovian analysis of the *small population* evolutionary game. In Section 8 we discuss the connections to learning theory. Finally, in Section 9 we summarize our results and present future research directions.

2 Preliminaries

2.1 "Classic" Centipede

The game of centipede has been introduced in [15]. In subsequent works, many variants of the original game have been introduced, but they all share some basic characteristics.

1. It is an extensive form game of perect information (a tree game).

2. The game tree has the "centipede shape", namely it consists of a path of M vertices, with each path vertex also having an additional edge ending at a leaf vertex. Hence, the game tree contains a total of $2M$ vertices; we denote the path vertices as $1, 2, ..., M$ and the leaves as $M+1, M+2, ..., 2M$.

3. The terminal vertices are $M+1$, $M+2$, ..., $2M$; the $(M+m)$-th vertex has payoffs (a_m, b_m).

4. The payoffs satisfy:

$$a_1 > a_2, a_3 > a_4, a_5 > a_6, ...$$
$$b_1, b_2 > b_3, b_4 > b_5, b_6 > b_7, ...$$

The game is played as follows:

1. P_1 plays at odd turns, P_2 plays at even turns.
2. The sum of payoffs at the t-th turn is $c_t = a_t + b_t = f(t)$, where f is an increasing function of t.
3. The c_t is divided as follows: if the moving player goes to the leaf vertex, he receives pc_t and the other player receives $(1-p)c_t$, where $p \in (\frac{1}{2}, 1]$.

In this paper we will always take $f(t) = t$. As an example, taking $M = 4$ and $p = \frac{3}{4}$, we will have

t	1	2	3	4
c_t	1.00	2.00	3.00	4.00
a_t	0.75	0.50	2.25	1.00
b_t	0.25	1.50	0.75	3.00

resulting in the game tree illustrated in Fig.1.

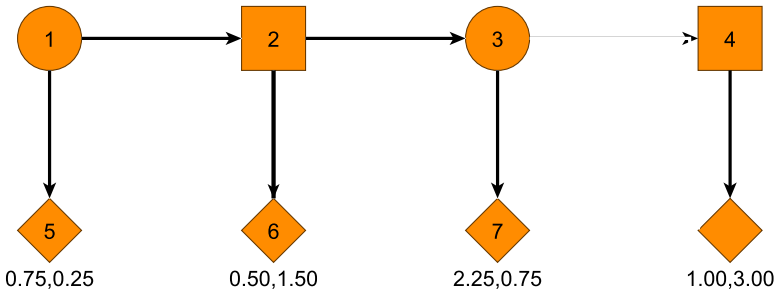


Figure 1: A Centipede Game

Hence at each nonterminal vertex, the player who has the move can decide to either terminate the game (move "down") or continue to the next stage (move "right"); this continues until a terminal (diamond) vertex is reached, at which point the game terminates unconditionally. In other words, whenever a player has the move he can choose to: (a) either get immediate payoff, (b) or continue the game hoping for a higher payoff at a later stage. Note that the *total* payoff increases with the length of the game and also that the terminating player receives a higher proportion of the total payoff.

Solving the game by *backward induction* we obtain the unique subgame perfect equilibrium. This, in the example of Fig. 1, is for the first player to immediately move down, resulting to payoff vector $(0.75, 0.25)$. However this is counterintuitive: if both players keep moving right, they will end up with payoffs $(1.00, 3.00)$. I.e., they would both benefit by using the strategy of always moving to the right; this however is not an equilibrium.

Let us now introduce a family of *strategies* $\rho_1, ..., \rho_M$ which can be used by both players. These strategies are defined as follows:

$$\forall m : \rho_m = \text{"I terminate the game as soon as it reaches or passes the } m\text{-th stage and it is my turn to play."}$$

Let us denote the two players by P_1 and P_2 and define for $i \in \{1,2\}$ and $m,n \in \{1,...,M\}$ the following:

$$Q_i(\rho_m, \rho_n) = \text{“Payoff to } P_i \text{ when } P_1 \text{ uses } \rho_m \text{ and } P_2 \text{ uses } \rho_n\text{”}.$$

Now let us compute some examples $Q_1(\rho_m, \rho_n)$, $Q_2(\rho_m, \rho_n)$, for the example game of Fig. 1.

1. Suppose that both players use ρ_1. Hence P_1 terminates the game at $t = 1$; the total payoff is $Q(\sigma_1, \sigma_2) = 1$ and it is split between the two players as follows: $Q_1(\rho_1, \rho_1) = \frac{3}{4} \cdot 1 = 0.75$ and $Q_2(\rho_1, \rho_1) = \frac{1}{4} \cdot 1 = 0.25$.

2. Suppose that P_1 uses ρ_2 and P_2 uses ρ_4. At $t = 2$, P_1's threshold is reached but he does not have the move, while P_2's threshold has not been yet reached; at $t = 3$, P_1's threshold is exceeded and he has the move (while P_2's threshold has not been reached yet) hence P_1 terminates the game. The total payoff is $Q(\rho_2, \rho_4) = 3$ and it is split between the two players as follows: $Q_1(\rho_2, \rho_4) = \frac{3}{4} \cdot 3 = 2.25$ and $Q_2(\rho_2, \rho_4) = \frac{1}{4} \cdot 3 = 0.75$.

3. Suppose that P_1 uses ρ_4 and P_2 uses ρ_3. At $t = 3$, P_1's threshold has not been reached, while P_2's threshold has been reached but he does not have the move; at $t = 4$, P_1's threshold has been reached and P_2's threshold is exceeded, hence P_2 (having the move) terminates the game. The total payoff is $Q(\rho_2, \rho_4) = 4$ and it is split between the two players as follows: $Q_1(\rho_2, \rho_4) = \frac{1}{4} \cdot 4 = 1.00$ and $Q_2(\rho_2, \rho_4) = \frac{3}{4} \cdot 4 = 3.00$.

2.2 One-Round Symmetric Centipede

We can employ a standard procedure to obtain, from the tree Centipede, a normal form bimatrix game, which we will call the *One-Round Symmetric Centipede* (OSC) game. We start by keeping the assumptions of the tree Centipede (total payoff at t is t, P_1 receives pt, P_2 receives $(1-p)t$). Further, we define $M \times M$ matrices $A^{(1)}$ and $A^{(2)}$ as follows:

$$\forall i \in \{1,2\} : A^{(i)}_{mn} = \text{“Payoff to } P_1 \text{ when: } P_1 \text{ plays } \rho_m,\ P_2 \text{ plays } \rho_n \text{ and } P_i \text{ plays first”}.$$

Now we suppose that P_1 plays first and we compute $A^{(1)}$. We reason as follows.

1. When i is odd and j is even:

$$A^{(1)}_{i,j} = \begin{cases} p\min(i,j), & \text{if } i < j, \\ (1-p)\min(i,j), & \text{if } i > j. \end{cases}$$

2. When both i and j are odd, $A^{(1)}_{i,j} = A^{(1)}_{i,j+1}$, since the second player will not play until we are in an even turn and the turn is equal or above j.

3. Similarly, when both i and j are even, $A^{(1)}_{i,j} = A^{(1)}_{i+1,j}$.

4. Finally, when i is even and j is odd, $A^{(1)}_{i,j} = A^{(1)}_{i+1,j+1}$.

Similar properties are also true for $A^{(2)}_{i,j}$. Letting A be the expected payoff matrix of OSC, we have

$$A = \frac{1}{2}\left(A^{(1)} + A^{(2)}\right).$$

Dropping the $\frac{1}{2}$ factor (which will not influence our later results), we more specifically have:

$$\begin{aligned}
\text{for } i \text{ odd, } j \text{ even: } & A_{i,j} = A^{(1)}_{i,j} + A^{(2)}_{i+1,j+1},\\
\text{for } i \text{ even, } j \text{ odd: } & A_{i,j} = A^{(1)}_{i+1,j+1} + A^{(2)}_{i,j},\\
\text{for } i \text{ odd, } j \text{ odd: } & A_{i,j} = A^{(1)}_{i,j+1} + A^{(2)}_{i+1,j},\\
\text{for } i \text{ even, } j \text{ even: } & A_{i,j} = A^{(1)}_{i+1,j} + A^{(2)}_{i,j+1}.
\end{aligned}$$

Let us compute some particular values.

1. For $A_{i,i}$ we have:

$$\begin{aligned}
\text{for } i \text{ odd: } & A_{i,i} = A^{(1)}_{i,i+1} + A^{(2)}_{i+1,i} = ip + i(1-p) = i,\\
\text{for } i \text{ even: } & A_{i,i} = A_{i,i} = A^{(1)}_{i+1,i} + A^{(2)}_{i,i+1} = i(1-p) + ip = i.
\end{aligned}$$

2. For $A_{i,1}$ with $i > 1$ we have:

$$\begin{aligned}
\text{for } i \text{ odd: } & A_{i,1} = A^{(1)}_{i+1,2} + A^{(2)}_{i,1} = 2(1-p) + (1-p) = 3(1-p),\\
\text{for } i \text{ even: } & A_{i,1} = A^{(1)}_{i,2} + A^{(2)}_{i+1,1} = 2(1-p) + s(1-p) = 3(1-p).
\end{aligned}$$

3. Similarly, we can calculate that:

$$A_{1,i} = 3p.$$

4. For $A_{i,j}$ with $i > j$:

$$\begin{aligned}
\text{for } i \text{ odd, } j \text{ even: } & A_{i,j} = A^{(1)}_{i,j} + A^{(2)}_{i+1,j+1} = j(1-p) + (j+1)(1-p) = (2j+1)\cdot(1-p),\\
\text{for } i \text{ even, } j \text{ odd: } & A_{i,j} = A^{(1)}_{i+1,j+1} + A^{(2)}_{i,j} = (j+1)(1-p) + j(1-p) = (2j+1)\cdot(1-p),\\
\text{for } i \text{ odd, } j \text{ odd: } & A_{i,j} = A^{(1)}_{i,j+1} + A^{(2)}_{i+1,j} = i(1-p) + (j+1)(1-p) = (2j+1)(1-p),\\
\text{for } i \text{ even, } j \text{ even: } & A_{i,j} = A^{(1)}_{i+1,j} + A^{(2)}_{i,j+1} = (j+1)(1-p) + j(1-p) = (2j+1)(1-p).
\end{aligned}$$

5. Similarly, we can calculate that for $i < j$:

$$A_{i,j} = (2i+1)p.$$

So finally we have

$$A = \begin{bmatrix}
1 & 3p & 3p & 3p & ... & 3p \\
3\,(1-p) & 2 & 5p & 5p & ... & 5p \\
3\,(1-p) & 5\,(1-p) & 3 & 7p & ... & 7p \\
3\,(1-p) & 5\,(1-p) & 7\,(1-p) & 4 & ... & 9p \\
... & ... & ... & ... & ... & ... \\
3\,(1-p) & 5\,(1-p) & 7\,(1-p) & 9\,(1-p) & ... & M
\end{bmatrix}.$$

3 Iterated Symmetric Centipede

Now we introduce the *Iterated Symmetric Centipede* (ISC) game. It is obtained from OSC in the standard manner which is used in the study of iterated games; i.e., ISC consists in playing OSC for T rounds. What makes ISC qualitatively different from OSC is that a player can use strategies with *memory*. In other words, the OSC strategy used in the t-th round of ISC may depend on the moves of previous rounds. Hence ISC has the same relationship to OSC as the Iterated Prisoner's Dilemma (IPD) has to Prisoner's Dilemma (PD). In this paper we concentrate on three ISC strategies:

1. $\sigma_{\text{All-}M}$**:** In every OSC game play ρ_M.
2. $\sigma_{\text{All-1}}$: In every OSC game play ρ_1.
3. σ_G or *Grim***:** Play ρ_M at the first round and, after that, as long as the opponent has played ρ_M in all previous rounds; if at some round the opponent plays ρ_m with $m < M$, then play ρ_1 in all subsequent rounds.

This strategy ensemble allows us to explore the interplay between the fully cooperating strategy $\sigma_{\text{All-}M}$, the fully defecting strategy $\sigma_{\text{All-1}}$ and the strategy σ_G which starts with cooperation but punishes defections. The analogy between $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G and the IPD strategies All-C, All-D, Grim is obvious.

More precisely, we define an ISC game as follows: it is played between two players, each of which can play one of the pure strategies $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G. At the beginning, each player chooses one of the pure strategies and uses it in playing a sequence of T rounds of OSC. Referring to $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G as the first, second and third strategy, we compute the payoff matrix C as follows:

C_{ij} = "Payoff to a user of the i-th strategy whe playing against a user of the j-th strategy".

The C matrix is computed in terms of the A matrix of OSC game. Let us give some examples:

1. C_{11} is the payoff obtained by a $\sigma_{\text{All-}M}$ user playing against another $\sigma_{\text{All-}M}$ user; since he plays ρ_M against ρ_M for T rounds, his payoff equals TA_{MM} .
2. Similarly, C_{12} is the payoff obtained by a $\sigma_{\text{All-}M}$ user playing against a $\sigma_{\text{All-1}}$ user; since he plays ρ_M against ρ_1 for T rounds, his payoff equals to TA_{M1} .
3. Consider now C_{23}; this is the payoff of a σ_G user playing against a $\sigma_{\text{All-1}}$ user. In the first round, he plays ρ_M against ρ_1 and receives A_{M1}; in the remaining $T-1$ rounds he plays ρ_1 against ρ_1 and receives $(T-1)A_{11}$; so his total payoff is $A_{M1} + (T-1)A_{11}$.

The remaining entries of C are computed in similar manner, resulting in the following expression for C:

$$C = \begin{bmatrix} TA_{MM} & TA_{M1} & TA_{MM} \\ TA_{1M} & TA_{11} & A_{1M} + (T-1)A_{11} \\ TA_{MM} & A_{M1} + (T-1)A_{11} & TA_{MM} \end{bmatrix} = \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p + (T-1) \\ TM & 3(1-p) + (T-1) & TM \end{bmatrix}$$

Finally, the bimatrix ISC game is $D = \left(C, C^T\right)$.